

Overview of Dropout regularization

Subject: Dropout regularization

$$\tilde{h} = r \odot h, \quad r_i \sim \text{Bernoulli}(p)$$

1.

$$\min_{w \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \ell(X_i^T w - Y_i) + \lambda \left(\alpha \|w\|_1 + (1 - \alpha) \|w\|_2^2 \right), \quad \alpha \in [0, 1]$$

2.

$$\min_{w \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \ell(X_i^T w - Y_i) + \lambda \|w\|_1$$

3.

where V is an underlying loss function that describes the cost of predicting $f(x)$ when the label is y , such as the square loss or hinge loss; and λ is a parameter which controls the importance of the regularization term. $R(f)$ is typically chosen to impose a penalty on the complexity of f . Concrete notions of complexity used include restrictions for smoothness and bounds on the vector space norm. A theoretical justification for regularization is that it attempts to impose Occam's razor on the solution (as depicted in the figure above, where the green function, the simpler one, may be preferred). From a Bayesian point of view, many regularization techniques correspond to imposing certain prior distributions on model parameters. Regularization can serve multiple purposes, including learning simpler models, inducing models to be sparse and introducing group structure into the learning problem. The same idea arose in many fields of science. A simple form of regularization applied to integral equations (Tikhonov regularization) is essentially a trade-off between fitting the data and reducing a norm of the solution. More recently, non-linear regularization methods, including total variation regularization, have become popular.

4.

Tikhonov-regularized least squares The learning problem with the least squares loss function and Tikhonov regularization can be solved analytically. Written in matrix form, the optimal w is the one for which the gradient of the loss function with respect to w is 0.

5.

$$w_{k+1} = \text{prox}_{\gamma, R}(w_k - \gamma \nabla F(w_k))$$

6.

Regularizers for multitask learning In the case of multitask learning, T problems are considered simultaneously, each related in some way. The goal is to learn T functions, ideally borrowing strength from the relatedness of tasks, that have predictive power. This is equivalent to learning the matrix $W : T \times D$.

7.

L_1 regularization can occasionally produce non-unique solutions. A simple example is provided in the figure when the space of possible solutions lies on a 45 degree line. This can be problematic for certain applications, and is overcome by combining L_1 with L_2 regularization in elastic net regularization, which takes the following form:

8.

$$I[f_n] = \int_{X \times Y} V(f_n(x), y) \rho(x, y) dx dy$$

9.

Regularization in machine learning In machine learning, a key challenge is enabling models to accurately predict outcomes on unseen data, not just on familiar training data. Regularization is crucial for addressing overfitting—where a model memorizes training data details but cannot generalize to new data. The goal of regularization is to encourage models to learn the broader patterns within the data rather than memorizing it. Techniques like early stopping, L1 and L2 regularization, and dropout are designed to prevent overfitting and underfitting, thereby enhancing the model's ability to adapt to and perform well with new data, thus improving model generalization.

10.

$$\text{prox}_{\lambda, R, g}(w_g) = \begin{cases} (1 - \lambda \|w_g\|^2) w_g, & \text{if } \|w_g\|^2 > \lambda \\ 0, & \text{if } \|w_g\|^2 \leq \lambda \end{cases}$$

11.

$$\min_{f_u \in R} f_{TL} = \min_{f_u \in R} \{ f_{TL} + f_{LU} + f_{LU} + f_{LU} + f_{LU} \}$$

12.

$$R(w) = \inf \{ \sum_{g=1}^G \|w_g\|^2 : w = \sum_{g=1}^G \bar{w}_g \}$$

13.

Other uses of regularization in statistics and machine learning Bayesian learning methods make use of a prior probability that (usually) gives lower probability to more complex models. Well-known model

selection techniques include the Akaike information criterion (AIC), minimum description length (MDL), and the Bayesian information criterion (BIC). Alternative methods of controlling overfitting not involving regularization include cross-validation. Examples of applications of different methods of regularization to the linear model are:

14.

Proximal methods While the L_1 norm does not result in an NP-hard problem, the L_1 norm is convex but is not strictly differentiable due to the kink at $x = 0$. Subgradient methods which rely on the subderivative can be used to solve L_1 regularized learning problems. However, faster convergence can be achieved through proximal methods. For a problem $\min_{w \in H} F(w) + R(w)$ such that F is convex, continuous, differentiable, with Lipschitz continuous gradient (such as the least squares loss function), and R is convex, continuous, and proper, then the proximal method to solve the problem is as follows. First define the proximal operator

15.

Elastic net regularization tends to have a grouping effect, where correlated input features are assigned equal weights. Elastic net regularization is commonly used in practice and is implemented in many machine learning libraries.